# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 8 — MATHS TWO-CHAPTER TEST SET A

Chapters 9 & 10: **The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

### SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The legs of a right triangle are 8 cm and 15 cm. Its hypotenuse is: 1

**(a)** 16 cm **(b)** 23 cm **(c)** 18 cm **(d)** 17 cm

**2.** The legs of a right triangle are 6 cm and 8 cm. Its hypotenuse is: 1

**(a)** 10 cm **(b)** 11 cm **(c)** 14 cm **(d)** 9 cm

**3.** The hypotenuse of a right triangle is 17 cm and one leg is 8 cm. The other leg is: 1

**(a)** 14 cm **(b)** 16 cm **(c)** 9 cm **(d)** 15 cm

**4.** Is a triangle with sides 6, 8 and 10 a right triangle? 1

**(a)** Only if it is isosceles **(b)** Cannot say

**(c)** Yes **(d)** No

**5.** Is a triangle with sides 14, 48 and 50 a right triangle? 1

**(a)** Only if it is isosceles **(b)** Yes

**(c)** No **(d)** Cannot say

**6.** On a map, 1 cm represents 150 km. A distance of 3 cm on the map represents: 1

**(a)** 225 km **(b)** 900 km **(c)** 450 km **(d)** 600 km

**7.** ₹ 600 is shared among A, B and C in the ratio 1:2:3. B gets: 1

**(a)** ₹ 200 **(b)** ₹ 100 **(c)** ₹ 300 **(d)** ₹ 400

**8.** ₹ 840 is shared among A, B and C in the ratio 2:4:8. B gets: 1

**(a)** ₹ 120 **(b)** ₹ 240 **(c)** ₹ 280 **(d)** ₹ 480

**9.** 8 workers finish a job in 9 days. How many days will 12 workers take? 1

**(a)** 6 **(b)** 8 **(c)** 5 **(d)** 13.5

**10.** In a pie chart, a sector representing 25% of the data has a central angle of: 1

**(a)** 25° **(b)** 100° **(c)** 90° **(d)** 50°

### SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 10–15 words each.*

**11.** Find the area of a right triangle with hypotenuse 17 cm and one leg 8 cm. 2

**12.** Name the hypotenuse in a right triangle PQR right-angled at Q. 2

**13.** A car takes 5 hours at 60 km/h. How long at 75 km/h? 2

**14.** Find the central angle of a sector representing 35% of the data. 2

## SECTION — C (Short Answer Type) 4 × 3 = 12

*Answer in about 30–40 words each.*

**15.** A rectangular park measures 9 m by 12 m. A boy walks along the diagonal instead of two sides. How much distance does he save? 3

**16.** 15 workers can complete a task in 20 days. How many days will 25 workers take, and how many extra days if 15 had worked? 3

**17.** A rectangular park measures 20 m by 21 m. A boy walks along the diagonal instead of two sides. How much distance does he save? 3

OR

Using $(n,\ \frac{n^2-1}{2},\ \frac{n^2+1}{2})$ find the triplet for $n = 3$ and verify. 3

**18.** 24 workers can complete a task in 18 days. How many days will 36 workers take, and how many extra days if 24 had worked? 3

OR

10 workers can complete a task in 30 days. How many days will 12 workers take, and how many extra days if 10 had worked? 3

## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 80–100 words each.*

**19.** (i) A 17 m ladder reaches a window 15 m above ground. A second window is 8 m above ground; how far must the foot be moved? 5
(ii) In which direction?

OR

**Case study:** A kite string of 125 m is fully out and the kite is vertically above a point 75 m from the boy. 5
(i) Find the height of the kite.
(ii) If 25 m more string is released and the kite is at the same height, how far horizontally is the kite?
(iii) Name the triplet.

**20.** (i) A train 300 m long crosses a pole in 15 s. Find its speed. 5
(ii) At the same speed, time to cross a 450 m bridge?

OR

(i) Mix paints A and B in ratio 3 : 5. How much A is needed for 24 L total? 5
(ii) If extra 10 L B is added, find the new ratio.

## END OF QUESTION PAPER Total: 40 Marks